Endocrine Emergencies in ICU

1. Thyroid Storm (Thyrotoxic Crisis)

Triggers
- Infection, surgery, trauma, childbirth, DKA
- Iodine load (contrast, amiodarone), stopping antithyroid drugs, radioiodine

Diagnosis
- Clinical: fever, tachycardia/AF, heart failure, agitation → coma, vomiting, diarrhoea, jaundice
- **Burch–Wartofsky ≥ 45** = likely storm (25–44 impending)
- ↓ TSH, ↑ FT4/FT3: hormone levels do **not** separate storm from simple thyrotoxicosis

Treatment (in this order)
- **β-blocker:** propranolol 60–80 mg PO q4h (also ↓ T4→T3); esmolol if heart failure or asthma
- **Thionamide:** PTU 500–1000 mg load then 250 mg q4h (blocks T4→T3) or methimazole 20 mg q4–6h
- **Iodine** (SSKI / Lugol) **≥ 1 h after** the thionamide
- **Hydrocortisone** 100 mg IV q8h
- Cooling + paracetamol (**avoid aspirin**: frees T4), fluids, treat trigger; cholestyramine; plasmapheresis or thyroidectomy if refractory

Key point: Thionamide before iodine. PTU preferred in 1st trimester and in storm; methimazole otherwise (PTU hepatotoxicity).

2. Myxedema Coma (Severe Hypothyroidism)

Who and why
- Elderly women, winter; untreated or stopped levothyroxine
- Triggers: infection, cold, sedatives/opioids, surgery, amiodarone, lithium

Features
- ↓ consciousness, **hypothermia**, bradycardia, hypotension
- **Hypoventilation → hypercapnia**, hyponatraemia, hypoglycaemia
- Pericardial effusion, ileus, non-pitting oedema, ↑ CK
- Primary: ↑ TSH, ↓ FT4. Low/normal TSH → central: think pituitary + adrenal

Treatment
- **Hydrocortisone 100 mg IV q8h first** (until adrenal insufficiency excluded)
- **IV levothyroxine** 200–400 µg load, then 50–100 µg/day; lower dose in elderly or coronary disease
- ± IV T3 5–20 µg then 2.5–10 µg q8h (risk of arrhythmia)
- Ventilate for hypercapnia; **passive rewarming**; cautious fluids; correct Na slowly; avoid sedatives

Key point: Steroid before thyroxine. Hypothermia + hypercapnia + hyponatraemia in a drowsy elderly patient = myxedema coma until proven otherwise.

3. Adrenocortical Insufficiency & Adrenal Crisis

Types
- **Primary** (Addison: autoimmune, TB, adrenal haemorrhage – anticoagulation, meningococcaemia): ↓ cortisol + ↓ aldosterone → **↑ K**, ↓ Na, hyperpigmentation
- **Secondary** (most common cause = **stopping long-term steroids**; pituitary disease): aldosterone intact → no hyperkalaemia, no pigmentation

Crisis
- Shock refractory to fluids and vasopressors, vomiting, abdominal pain, fever, confusion
- Hyponatraemia, hypoglycaemia, eosinophilia, ↑ urea

Diagnosis
- Draw **cortisol + ACTH**, then treat at once (never wait)
- Random cortisol < 3 µg/dL (≈ 80 nmol/L) confirms; Synacthen 250 µg peak < 18 µg/dL (500 nmol/L) = deficient (lower cut-offs with newer assays)
- Dexamethasone does not interfere with the cortisol assay

Treatment
- **Hydrocortisone 100 mg IV**, then 200 mg/24 h (50 mg q6h or infusion)
- Saline 1 L in the first hour + glucose; fludrocortisone not needed while hydrocortisone ≥ 50 mg/day
- Taper when stable; sick-day rules and emergency card

Key point: Hyperkalaemia + pigmentation = primary. Refractory shock in a patient on long-term steroids = adrenal crisis.

4. Pheochromocytoma / Paraganglioma Crisis

Triggers
- **β-blocker without α-blockade**, anaesthesia, surgery or tumour manipulation
- Glucocorticoids, metoclopramide, tricyclics, opioids, IV contrast (rare)

Features
- Paroxysms of hypertension (or **shock**), headache, sweating, palpitations
- Catecholamine (Takotsubo) cardiomyopathy, pulmonary oedema, arrhythmia, lactic acidosis, hyperglycaemia, multi-organ failure

Diagnosis
- **Plasma free metanephrines** (most sensitive) or 24-h urine fractionated metanephrines
- Then CT/MRI; functional imaging (Ga-68 DOTATATE PET)

Treatment
- **α first:** phentolamine 1–5 mg IV boluses/infusion; nicardipine, clevidipine, nitroprusside, magnesium
- **β-blocker only after α-blockade** (unopposed α → hypertensive crisis)
- Shock: fluids, inotropes; ECMO in catecholamine cardiomyopathy
- Pre-op: phenoxybenzamine or doxazosin 10–14 days + salt/fluids; post-op watch **hypotension and hypoglycaemia**

Key point: Never β before α. Unexplained shock or cardiomyopathy at induction of anaesthesia → think pheochromocytoma.

5. Hypercalcemic Crisis

Definition and causes
- Severe: Ca > 14 mg/dL (3.5 mmol/L); corrected Ca = Ca + 0.8 × (4 − albumin g/dL)
- **Primary hyperparathyroidism + malignancy = 90%**
- Malignancy: PTHrP (squamous), osteolytic (myeloma, breast), calcitriol (lymphoma)
- Others: vitamin D toxicity, granulomas (sarcoid, TB), thiazides, lithium, milk-alkali, immobilisation

Features
- Stones, bones, groans, moans: polyuria, dehydration, AKI, confusion, constipation, pancreatitis
- ECG: **short QT**, bradyarrhythmia

Treatment
- **Saline** 200–300 mL/h → urine 100–150 mL/h
- **Calcitonin** 4 IU/kg q12h: fast (hours), tachyphylaxis in 48 h
- **Zoledronic acid** 4 mg IV: effect in 2–4 days
- **Denosumab** 120 mg SC if renal failure or refractory (risk of hypocalcaemia)
- Loop diuretic only after volume repletion; **steroids** for vitamin D excess, granulomas, lymphoma
- **Dialysis** (low-Ca bath) if AKI or heart failure; parathyroidectomy for primary HPT

Key point: Fluids + calcitonin for speed, bisphosphonate for duration. PTH high = parathyroid; PTH suppressed = look for malignancy.

6. Hypocalcemic Crisis & Hypoparathyroidism

Causes
- Post-thyroid/parathyroid surgery (incl. **hungry bone**), **hypomagnesaemia**
- Sepsis, pancreatitis, massive transfusion (citrate), rhabdomyolysis, tumour lysis, hyperphosphataemia
- Vitamin D deficiency, bisphosphonate/denosumab; alkalosis ↓ ionised Ca

Features
- Perioral and finger paraesthesia, tetany, **Chvostek**, **Trousseau**
- Laryngospasm, bronchospasm, seizures; **long QT**, hypotension, heart failure
- Measure **ionised Ca** (normal 1.1–1.3 mmol/L)

Treatment
- Symptoms or long QT: **Ca gluconate 10% 10–20 mL** IV over 10 min, then infusion (0.5–1.5 mg/kg/h elemental)
- Ca chloride has 3× the elemental Ca: central line (tissue necrosis)
- **Correct magnesium** (refractory otherwise); do not mix with bicarbonate; care with digoxin
- Hypoparathyroidism: oral calcium + calcitriol; keep Ca low-normal (avoid hypercalciuria)

Key point: Hypocalcaemia that will not correct = check magnesium.

Endocrine Emergencies in ICU

7. SIADH & Hyponatremia in the ICU

Approach

- Serum osm low (hypotonic) → urine osm > 100 (ADH acting) → urine Na > 30 + euvolaemia → SIADH
- Exclude hypothyroidism, adrenal insufficiency, diuretics
- Causes: CNS, lung disease, small cell lung cancer, drugs (SSRIs, carbamazepine, cyclophosphamide, opioids), pain, nausea, post-op

Severe symptoms (seizure, coma)

- **3% NaCl 150 mL over 20 min**, repeat up to 2–3 times until Na ↑ 4–6 mmol/L (bolus approach, SALSA 2021)
- Limit: **≤ 10 mmol/L in 24 h** (≤ 8 if high ODS risk: Na < 105, alcohol, malnutrition, hypokalaemia, liver disease)
- Overcorrection: D5W ± desmopressin 2 µg IV

Chronic SIADH

- Fluid restriction < 800 mL/day; **oral urea** 15–30 g/day
- **Tolvaptan**: start in hospital, no fluid restriction at the start, avoid in liver disease

Cerebral salt wasting

- SAH/neurosurgery, **hypovolaemic** with high urine Na → give saline ± fludrocortisone (never fluid restrict)

Key point: Volume status separates SIADH (euvolaemic) from cerebral salt wasting (hypovolaemic).

8. Diabetes Insipidus (Central & Nephrogenic)

Recognition

- Polyuria > 3 L/day (> 50 mL/kg/day), **urine osm < 300** (often < 200), urine SG < 1.005
- Hypernatraemia when water intake is limited (ICU, sedated patients)

Causes

- **Central:** TBI, pituitary surgery (**triphasic**: DI → SIADH → DI), brain death, tumours, sarcoid, meningitis
- **Nephrogenic: lithium**, hypercalcaemia, hypokalaemia, demeclocycline, amphotericin, post-obstruction, ATN recovery

Diagnosis

- Desmopressin test: urine osm ↑ > 50% → central; < 10% → nephrogenic
- Copeptin: baseline > 21.4 pmol/L → nephrogenic; stimulated < 4.9 → central

Treatment

- Replace free water (enteral water or D5W); chronic hypernatraemia: ↓ Na ≤ 10–12 mmol/L/day
- **Central: desmopressin** 1–2 µg IV/SC q12h (watch for hyponatraemia)
- **Nephrogenic:** stop the cause, **thiazide** + low-salt diet, amiloride for lithium, NSAIDs
- Brain-dead donor: vasopressin or desmopressin

Key point: High urine output + high Na + dilute urine = DI; the response to desmopressin tells central from nephrogenic.

9. Hypopituitarism & Pituitary Apoplexy

Pituitary apoplexy

- Haemorrhage/infarction of a pituitary adenoma: **sudden severe headache**, visual loss (bitemporal), ophthalmoplegia (**CN III** most often), ↓ consciousness
- Triggers: anticoagulation, cardiac surgery, dopamine agonists, pregnancy, dynamic pituitary tests
- **Sheehan:** postpartum haemorrhage → failure of lactation, then hypopituitarism

Diagnosis

- **MRI** (better than CT); draw cortisol + pituitary hormones
- Pattern: low cortisol with low/normal ACTH; low FT4 with low/normal TSH; hyponatraemia without hyperkalaemia

Treatment

- **Hydrocortisone 100 mg IV at once**, then 50 mg q6h
- Urgent neurosurgery if severe or worsening visual loss or ↓ GCS; otherwise conservative
- Replace **cortisol before thyroxine**; watch for DI and SIADH

Key point: Thunderclap headache + visual loss + shock = pituitary apoplexy: give hydrocortisone before imaging results.

10. Carcinoid Crisis

Mechanism

- Neuroendocrine tumour (often midgut with **liver metastases**) releases serotonin, histamine, kallikrein
- Carcinoid heart disease: right-sided valves (tricuspid regurgitation, pulmonary stenosis)

Features

- Flushing, **bronchospasm**, profound **hypotension** (sometimes hypertension), arrhythmia, diarrhoea, hyperthermia

Triggers

- Anaesthesia induction, surgery or tumour manipulation, embolisation, PRRT
- **Catecholamines** (adrenaline, ephedrine) and histamine-releasing drugs (morphine, atracurium)

Diagnosis and treatment

- 24-h urine **5-HIAA**, chromogranin A
- **Octreotide** 100–500 µg IV bolus, then 50–200 µg/h; prophylaxis before procedures
- Hypotension: fluids, **vasopressin or phenylephrine**; avoid catecholamines (worsen mediator release)
- Bronchospasm: octreotide, ipratropium, antihistamine, steroid; avoid β-agonists if possible

Key point: Hypotension in carcinoid crisis = octreotide, not adrenaline.

11. ICU Exam Pearls: Master List for Endocrine

- Thyroid storm: **PTU/methimazole before iodine**; avoid aspirin
- Myxedema coma and apoplexy: **hydrocortisone before thyroxine**
- Adrenal crisis: **treat first, test later**; dexamethasone does not affect cortisol assay
- Primary AI: ↑ K + pigmentation; secondary: neither
- Pheochromocytoma: **α before β**; plasma free metanephrines
- Hypercalcaemia: saline + calcitonin (fast), zoledronate (2–4 days), denosumab in renal failure
- Hypocalcaemia refractory → **magnesium**
- SIADH: euvolaemic, urine osm > 100, urine Na > 30; CSW: hypovolaemic
- Hyponatraemia correction ≤ 10 mmol/L/24 h (≤ 8 if high risk)
- DI: dilute urine + hypernatraemia; desmopressin response = central
- Carcinoid crisis: octreotide, avoid catecholamines
- Sick euthyroid in ICU: low T3 (± low T4) with normal/low TSH; do not treat

Key point: Most endocrine crises: start empiric treatment after drawing bloods; do not wait for results.

12. ICU Exam Pearls: High-Yield Numbers

- Burch–Wartofsky ≥ 45 = thyroid storm; 25–44 = impending
- Hydrocortisone 100 mg IV then 200 mg/day (adrenal crisis, apoplexy, myxedema)
- Random cortisol < 3 µg/dL = adrenal insufficiency; Synacthen peak < 18 µg/dL abnormal
- Severe hypercalcaemia: Ca > 14 mg/dL (3.5 mmol/L); calcitonin 4 IU/kg; zoledronate 4 mg
- Ionised Ca normal 1.1–1.3 mmol/L; Ca gluconate 10% 10–20 mL over 10 min
- Severe hyponatraemia: 3% NaCl 150 mL over 20 min ×2–3; target ↑ 4–6 mmol/L
- ODS risk: Na < 105, alcohol, malnutrition, hypokalaemia, liver disease
- DI: urine > 3 L/day, urine osm < 300; desmopressin 1–2 µg IV
- Copeptin > 21.4 pmol/L = nephrogenic DI
- Octreotide 100–500 µg bolus then 50–200 µg/h
- Phentolamine 1–5 mg IV boluses for catecholamine crisis
- IV levothyroxine 200–400 µg load in myxedema coma

Key point: Learn the doses and cut-offs: they are the commonest MCQ distractors in this chapter.

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Triggers

- Infection, surgery, trauma, childbirth, DKA
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Treatment (in this order)

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Key point: Thionamide before iodine. PTU preferred in 1st trimester and in storm; methimazole otherwise (PTU hepatotoxicity).

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Who and why

- Elderly women, winter; untreated or stopped levothyroxine
- Triggers: infection, cold, sedatives/opioids, surgery, amiodarone, lithium

Features

- ↓ consciousness, **hypothermia**, bradycardia, hypotension
- **Hypoventilation → hypercapnia**, hyponatraemia, hypoglycaemia
- Pericardial effusion, ileus, non-pitting oedema, ↑ CK
- Primary: ↑ TSH, ↓ FT4. Low/normal TSH → central: think pituitary + adrenal

Treatment

- **Hydrocortisone 100 mg IV q8h first** (until adrenal insufficiency excluded)
- **IV levothyroxine** 200–400 µg load, then 50–100 µg/day; lower dose in elderly or coronary disease
- ± IV T3 5–20 µg then 2.5–10 µg q8h (risk of arrhythmia)
- Ventilate for hypercapnia; **passive rewarming**; cautious fluids; correct Na slowly; avoid sedatives

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- Shock refractory to fluids and vasopressors, vomiting, abdominal pain, fever, confusion
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Diagnosis

- Draw **cortisol + ACTH**, then treat at once (never wait)
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Treatment

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- Taper when stable; sick-day rules and emergency card

Key point: Hyperkalaemia + pigmentation = primary. Refractory shock in a patient on long-term steroids = adrenal crisis.

4. Pheochromocytoma / Paraganglioma Crisis

Triggers

- **β-blocker without α-blockade**, anaesthesia, surgery or tumour manipulation
- Glucocorticoids, metoclopramide, tricyclics, opioids, IV contrast (rare)

Features

- Paroxysms of hypertension (or **shock**), headache, sweating, palpitations
- Catecholamine (Takotsubo) cardiomyopathy, pulmonary oedema, arrhythmia, lactic acidosis, hyperglycaemia, multi-organ failure

Diagnosis

- **Plasma free metanephrines** (most sensitive) or 24-h urine fractionated metanephrines
- Then CT/MRI; functional imaging (Ga-68 DOTATATE PET)

Treatment

- **α first:** phentolamine 1–5 mg IV boluses/infusion; nicardipine, clevidipine, nitroprusside, magnesium
- **β-blocker only after α-blockade** (unopposed α → hypertensive crisis)
- Shock: fluids, inotropes; ECMO in catecholamine cardiomyopathy
- Pre-op: phenoxybenzamine or doxazosin 10–14 days + salt/fluids; post-op watch **hypotension and hypoglycaemia**

Key point: Never β before α. Unexplained shock or cardiomyopathy at induction of anaesthesia → think pheochromocytoma.

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Definition and causes

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- Others: vitamin D toxicity, granulomas (sarcoid, TB), thiazides, lithium, milk-alkali, immobilisation

Features

- Stones, bones, groans, moans: polyuria, dehydration, AKI, confusion, constipation, pancreatitis
- ECG: **short QT**, bradyarrhythmia

Treatment

- **Saline** 200–300 mL/h → urine 100–150 mL/h
- **Calcitonin** 4 IU/kg q12h: fast (hours), tachyphylaxis in 48 h
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- **Dialysis** (low-Ca bath) if AKI or heart failure; parathyroidectomy for primary HPT

Key point: Fluids + calcitonin for speed, bisphosphonate for duration. PTH high = parathyroid; PTH suppressed = look for malignancy.

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Causes

- Post-thyroid/parathyroid surgery (incl. **hungry bone**), **hypomagnesaemia**
- Sepsis, pancreatitis, massive transfusion (citrate), rhabdomyolysis, tumour lysis, hyperphosphataemia
- Vitamin D deficiency, bisphosphonate/denosumab; alkalosis ↓ ionised Ca

Features

- Perioral and finger paraesthesia, tetany, **Chvostek**, **Trousseau**
- Laryngospasm, bronchospasm, seizures; **long QT**, hypotension, heart failure
- Measure **ionised Ca** (normal 1.1–1.3 mmol/L)

Treatment

- Symptoms or long QT: **Ca gluconate 10% 10–20 mL** IV over 10 min, then infusion (0.5–1.5 mg/kg/h elemental)
- Ca chloride has 3× the elemental Ca: central line (tissue necrosis)
- **Correct magnesium** (refractory otherwise); do not mix with bicarbonate; care with digoxin
- Hypoparathyroidism: oral calcium + calcitriol; keep Ca low-normal (avoid hypercalciuria)

Key point: Hypocalcaemia that will not correct = check magnesium.

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Approach

- Serum osm low (hypotonic) → urine osm > 100 (ADH acting) → urine Na > 30 + euvolaemia → SIADH
- Exclude hypothyroidism, adrenal insufficiency, diuretics
- Causes: CNS, lung disease, small cell lung cancer, drugs (SSRIs, carbamazepine, cyclophosphamide, opioids), pain, nausea, post-op

Severe symptoms (seizure, coma)

- **3% NaCl 150 mL over 20 min**, repeat up to 2–3 times until Na ↑ 4–6 mmol/L (bolus approach, SALSA 2021)
- Limit: **≤ 10 mmol/L in 24 h** (≤ 8 if high ODS risk: Na < 105, alcohol, malnutrition, hypokalaemia, liver disease)
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Chronic SIADH

- Fluid restriction < 800 mL/day; **oral urea** 15–30 g/day
- **Tolvaptan**: start in hospital, no fluid restriction at the start, avoid in liver disease

Cerebral salt wasting

- SAH/neurosurgery, **hypovolaemic** with high urine Na → give saline ± fludrocortisone (never fluid restrict)

Key point: Volume status separates SIADH (euvolaemic) from cerebral salt wasting (hypovolaemic).

8. Diabetes Insipidus (Central & Nephrogenic)

Recognition

- Polyuria > 3 L/day (> 50 mL/kg/day), **urine osm < 300** (often < 200), urine SG < 1.005
- Hypernatraemia when water intake is limited (ICU, sedated patients)

Causes

- **Central:** TBI, pituitary surgery (**triphasic**: DI → SIADH → DI), brain death, tumours, sarcoid, meningitis
- **Nephrogenic: lithium**, hypercalcaemia, hypokalaemia, demeclocycline, amphotericin, post-obstruction, ATN recovery

Diagnosis

- Desmopressin test: urine osm ↑ > 50% → central; < 10% → nephrogenic
- Copeptin: baseline > 21.4 pmol/L → nephrogenic; stimulated < 4.9 → central

Treatment

- Replace free water (enteral water or D5W); chronic hypernatraemia: ↓ Na ≤ 10–12 mmol/L/day
- **Central: desmopressin** 1–2 µg IV/SC q12h (watch for hyponatraemia)
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- Brain-dead donor: vasopressin or desmopressin

Key point: High urine output + high Na + dilute urine = DI; the response to desmopressin tells central from nephrogenic.

9. Hypopituitarism & Pituitary Apoplexy

Pituitary apoplexy

- Haemorrhage/infarction of a pituitary adenoma: **sudden severe headache**, visual loss (bitemporal), ophthalmoplegia (**CN III** most often), ↓ consciousness
- Triggers: anticoagulation, cardiac surgery, dopamine agonists, pregnancy, dynamic pituitary tests
- **Sheehan:** postpartum haemorrhage → failure of lactation, then hypopituitarism

Diagnosis

- **MRI** (better than CT); draw cortisol + pituitary hormones
- Pattern: low cortisol with low/normal ACTH; low FT4 with low/normal TSH; hyponatraemia without hyperkalaemia

Treatment

- **Hydrocortisone 100 mg IV at once**, then 50 mg q6h
- Urgent neurosurgery if severe or worsening visual loss or ↓ GCS; otherwise conservative
- Replace **cortisol before thyroxine**; watch for DI and SIADH

Key point: Thunderclap headache + visual loss + shock = pituitary apoplexy: give hydrocortisone before imaging results.

10. Carcinoid Crisis

Mechanism

- Neuroendocrine tumour (often midgut with **liver metastases**) releases serotonin, histamine, kallikrein
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Features

- Flushing, **bronchospasm**, profound **hypotension** (sometimes hypertension), arrhythmia, diarrhoea, hyperthermia

Triggers

- Anaesthesia induction, surgery or tumour manipulation, embolisation, PRRT
- **Catecholamines** (adrenaline, ephedrine) and histamine-releasing drugs (morphine, atracurium)

Diagnosis and treatment

- 24-h urine **5-HIAA**, chromogranin A
- **Octreotide** 100–500 µg IV bolus, then 50–200 µg/h; prophylaxis before procedures
- Hypotension: fluids, **vasopressin or phenylephrine**; avoid catecholamines (worsen mediator release)
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Key point: Hypotension in carcinoid crisis = octreotide, not adrenaline.

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